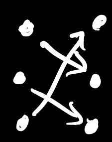
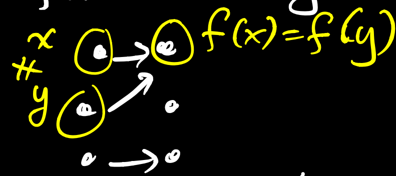
Lesson 21 One-to-one and Onto

Def $f: A \to B$

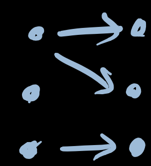
a) f is one-to-one (injective) if $f(x) = f(y)$ implies $x = y$ for all $x, y \in A$.



1-to-1



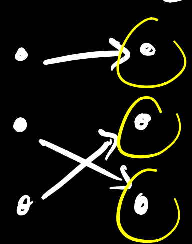
not 1-to-1



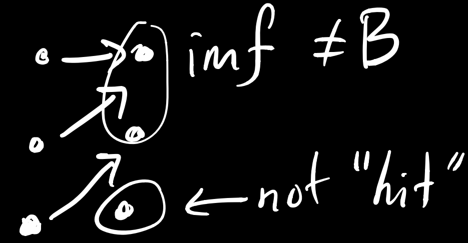
not even a function

b) f is onto (surjective) if for every $y \in B$, there is an element $x \in A$ s.t. $f(x) = y$.

Equivalently, $\operatorname{im} f = B$



onto



not onto

c) f is bijective (f is a bijection) if f is injective and surjective.

injective: 1-to-1, surjective: onto

Ex a) $i_A : A \to A$, $i_A(x) = x$ identity fn

1-to-1: If $\underbrace{i_A(x)}_{x} = \underbrace{i_A(y)}_{y}$, then $x = y$.

Onto: let $y \in A$ (codomain), then $y \in A$ (domain) and

$i_A(y) = y$.

domain codomain

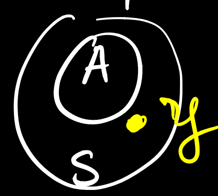
b) <u>inclusion</u> $i : A \to S$, $A \subseteq S$,

$i(x) = x$

1-to-1? Yes (same as (a))

<u>onto</u>? <u>No</u>, here is the counterexample:

Let $y \in S$ but $y \notin A$.



Then there is no $x \in A$ s.t.

$i(x) = y$, x would have to be y, but $y \notin$ domain

c) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x$

1-to-1? Yes

If $f(x) = f(y)$, then $\cancel{2}x = \cancel{2}y$ and $x = y$

onto? Yes

Let $y \in \mathbb{R}$ (codomain), then $x = \frac{y}{2} \in \mathbb{R}$ (domain) and $f(x) = f(\frac{y}{2}) = \cancel{2}(\frac{y}{\cancel{2}}) = y$

f is bijective (so is i_A)

d) $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x$

1-to-1? Yes (same as (c))

onto? No

Let $y = 1 \in \mathbb{Z}$ (codomain)

There is no $x \in \mathbb{Z}$ (domain) s.t. $f(x) = 1$. If there were, then

$f(x) = 2x = 1 \longrightarrow x = 1/2 \notin \mathbb{Z}$

Ex $f: \mathbb{Z} \to \mathbb{Z}$ $\quad f(x) = \begin{cases} x/2, & \cancel{x}\ x \text{ even} \\ \frac{x+1}{2}, & x \text{ odd} \end{cases}$

- 1-to-1? No, $\underbrace{f(1)}_{\frac{1+1}{2}} = \underbrace{f(2)}_{2/2} = 1$

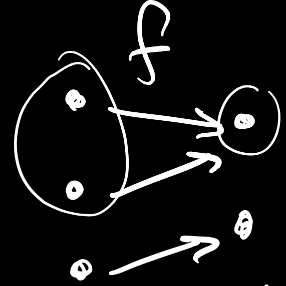
- onto? Yes
  Let $y \in \mathbb{Z}$ (codomain), then $x = 2y \in \mathbb{Z}$ (domain) and x is even, so $f(x) = f(2y) = \frac{2y}{2} = y$

Ex $f: \underbrace{\mathbb{R} \times \mathbb{R}}_{(x,y)} \to \mathbb{R}$, $f(x,y) = x + y$

1-to-1? No, $1 + 1 = 2 + 0 = 2$, but $(1,1) \neq (2,0)$

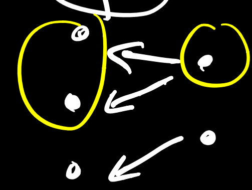
onto? Yes.
Let $y \in \mathbb{R}$ (codomain), then $\underbrace{(y,0) \in \mathbb{R} \times \mathbb{R}}_{\text{(domain)}}$ and $f(y,0) = y + 0 = y$.

Inverse of a function



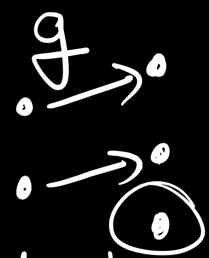
not 1-to-1

f^{-1} fn is also relation



f^{-1} is a relation (inverse relation of f)

f^{-1} is NOT fn



not onto

g^{-1} is a relation (inverse rel of g)

g^{-1} is not a fn

Thm Let $f: A \to B$. Then the inverse relation f^{-1} from B to A is a function, if and only if f is 1-to-1 and onto i.e. f is a bijection.

Proof ($\Rightarrow$) (If inverse relation f^{-1} is a fn then f is bijective)

First we show that f is 1-to-1.
Suppose $f(x) = f(y) = z$. Then $(x, z) \in f$ and $(y, z) \in f$. So, $(z, x) \in f^{-1}$ and $(z, y) \in f^{-1}$.
Since f^{-1} is a function, we have $x = y$.

Next, we show that f is onto.

Let $y \in B$. Since f^{-1} is a function (from B to A)

$f^{-1}(y) = x$ for some $x \in A$, i.e.

$(y, x) \in f^{-1}$, which implies $(x, y) \in f$,

i.e. $y = f(x)$.

($\Leftarrow$) (If f is a $\overbrace{\text{bijection}}^{\text{one-to one \& onto from } B \text{ to } A}$, then f^{-1} is a fn)

Let $y \in B$.

Since f is onto, we have $y = f(x)$ for some $x \in A$. So $(x, y) \in f$ and $(y, x) \in f^{-1}$,

i.e. $x = f^{-1}(y)$

Suppose $(x, y) \in f^{-1}$ and $(x, z) \in f^{-1}$

then $(y, x) \in f$ and $(z, x) \in f$.

Since f is one-to-one, $y = z$.

So f^{-1} is a fn.